

number of missing jobs below MW , given by $\Delta b = Emp^1[w < MW] - Emp^0[w < MW]$, reflects the bite of the minimum wage.² Here $Emp^1[\cdot]$ and $Emp^0[\cdot]$ are the actual and counterfactual frequency distributions of wages, respectively.

Not all missing jobs below the minimum wage are destroyed. Some or all of the jobs below the minimum wage may be preserved—with their hourly pay raised to the minimum wage, creating a spike at MW . Some jobs may be pushed slightly above the minimum wage to maintain wage hierarchy within the firm or because the minimum wage raises the bargaining power of workers (e.g., [Flinn 2011](#)). Moreover, a minimum wage increase might induce low-wage workers to participate in job search, some of whom may find a job above the minimum wage. However, the ripple effects of the minimum wage are likely to fade out at a certain point, which we denote $\bar{W}$ in [Figure I](#). In models with labor market frictions, wage spillovers also typically fade out, because workers and firms in the upper tail of the wage distribution are operating in different labor market segments (see [Van den Berg and Ridder 1998](#); [Engbom and Moser 2017](#) for examples of such models).

The neoclassical model suggests that there may be some positive employment effects in the upper tail of the wage distribution caused by labor-labor substitution. However, as we discuss in [Online Appendix B](#), because minimum wage workers' share in overall production is very small (around 2% in the United States), reasonable calibrations of a neoclassical model would suggest very small upper-tail effects. For example, if we consider an elasticity of substitution between high- and low-wage workers of around 1.4 based on [Katz and Murphy \(1992\)](#), and an output demand elasticity of around 1 based on [Aaronson and French \(2007\)](#), the implied upper-tail employment elasticity with respect to the minimum wage would be around 0.006. In [Online Appendix Table B.1](#), we show that reasonable variations in the key parameters uniformly suggest that plausible estimates of minimum wage impact on upper-tail employment should be very small. Moreover, any

2. When we refer to the “bite” of the minimum wage, or to the extent to which the minimum wage is “binding,” we mean how effective the minimum wage is in raising wages at the bottom. Therefore, the bite is a function of (i) how many workers are earning below the new minimum wage, (ii) how many of those workers are legally covered by the policy, and (iii) the extent of compliance.

theoretical upper-tail effects would be positive, so ignoring them will overstate the net job losses.

We assess the employment effect of the minimum wage on low-wage workers by summing the missing and excess jobs, $\Delta b + \Delta a$, which is equal to the employment change below a wage threshold $\bar{W}$: $\Delta b + \Delta a = \text{Emp}^1[w < \bar{W}] - \text{Emp}^0[w < \bar{W}]$. Such an estimator is similar to the “bunching” method developed in the recent public finance literature, which uses bunching around points that feature discontinuities in incentives to elicit behavioral responses (Kleven 2016). Although the estimation of the overall effect on low-wage jobs does not require a decomposition by wage bins relative to the minimum (e.g., into excess and missing jobs), such a decomposition does help assess the bite of the policy and exactly how the policy affects jobs and wages at the bottom. For example, the shape of the excess jobs can tell us about the extent of wage spillovers.

II.B. Empirical Implementation

The key empirical challenge is to estimate the counterfactual wage frequency distribution in the absence of a minimum wage increase. Instead of using either ad hoc functional forms (Meyer and Wise 1983; Dickens, Machin, and Manning 1998) or the distribution prior to the minimum wage increase (Harasztosi and Lindner forthcoming), we exploit state-level variation in the minimum wage and identify the counterfactual distribution using a difference-in-differences event study design. Our event-based approach uses a similar framework as Autor, Donohue, and Schwab (2006) and examines employment changes within an eight-year window around 138 prominent state-level minimum wage events, where states increased their minimum wage by at least \$0.25, and where at least 2% of the workers were directly affected by the increase.³ By focusing on employment changes around the event window, we incompletely capture long-run effects of the minimum wage. Nevertheless, as we show below, we find no evidence of a change in employment up to five years after the minimum wage hike, and so it strikes us as unlikely that our empirical

3. We exclude federal increases from our primary sample of events because for these events, the change in missing jobs, Δb , is identified only from time-series variation—as there are no “control states” with a wage floor lower than the new minimum wage. However, we show in Online Appendix Table A.4 that our employment and wage estimates are similar when we include federal events as well.

design misses important long-term employment changes. [Online Appendix Table A.6](#) shows the robustness of estimates to alternative window lengths, including allowing for up to a seven-year post-treatment period.

We estimate the effect of the minimum wage not just on aggregate employment but also on employment in every \$0.25 wage bin. Our basic regression specification is the following:

$$(1) \quad \frac{E_{sjt}}{N_{st}} = \sum_{\tau=-3}^4 \sum_{k=-4}^{17} \alpha_{\tau k} I_{sjt}^{\tau k} + \mu_{sj} + \rho_{jt} + \Omega_{sjt} + u_{sjt},$$

where E_{sjt} is the employment in \$0.25 wage bins j in state s and at quarter t , and N_{st} is the size of the population in state s and quarter t . The treatment dummy $I_{sjt}^{\tau k}$ equals 1 if the minimum wage was raised τ years from date t and for the \$0.25 wage bins j that fall between k and $k + 1$ dollars relative to the new minimum wage. This definition implies that $\tau = 0$ represents the first year following the minimum wage increase (i.e., the quarter of treatment and the subsequent three quarters), and $\tau = -1$ is the year (four quarters) prior to treatment. Moreover, the $I_{sjt}^{\tau k}$ treatment variables are a function of not only state and time but also the wage bins. For instance, $k = 0$ represents the four \$0.25 bins between MW and $MW + \$0.99$ and $k = -1$ is a “below” bin with wages paying between $MW - \$0.01$ and $MW - \$1.00$. Our benchmark specification also controls for state-by-wage-bin and period-by-wage-bin effects, μ_{sj} and ρ_{jt} . This allows us to control for state-specific factors in the earnings distribution and also the nationwide evolution of wage inequality. Finally, Ω_{sjt} include controls for small or federal increases, and u_{sjt} is the error term.⁴ We cluster our standard errors by state, the level at which policy is assigned. Our standard errors therefore account for the possibility that employment changes at different parts of the wage distribution may be correlated within a state.

4. Our primary minimum wage events exclude very small increases. To ensure they do not confound our main effects, we include controls for these small events. We also separately control for federal minimum wages. In particular, separately for small events and federal events, we construct a set of six variables by interacting $\{BELOW, ABOVE\} \times \{EARLY, PRE, POST\}$. Here *BELOW* and *ABOVE* are dummies equal to 1 for all wage bins that are within \$4 below and above the new minimum, respectively; *EARLY*, *PRE*, and *POST* are dummies that take on 1 if $-3 \leq \tau \leq -2$, $\tau = -1$, or $0 \leq \tau \leq 4$, respectively. These two sets of six variables are included as controls in the regression (Ω_{sjt} in [equation \(1\)](#)).

As with any difference-in-differences design our approach identifies the causal effect of the minimum wage under the assumption that the entire frequency distribution of wages in the treated and untreated states would move in parallel in the absence of the policy change. Although this assumption cannot be tested directly, we conduct a variety of checks whose results will be reported below. As is standard, we use the leading terms to assess preexisting trends. As an added check, when we calculate event-by-event estimates in [Section III.C](#), we test whether the distribution of leading effects is consistent with a sharp null of zero effects everywhere.

In addition, since our approach locates the source of the employment effects within the wage distribution, we can use the upper-tail employment changes as an added falsification test.⁵ Because large positive or negative changes in jobs paying above, say, \$15 an hour are unlikely to reflect the causal effect of the minimum wage, reporting such employment changes in the upper tail can be highly informative about model validity. Moreover, the potential bias from the confounding factors affecting the upper tail can be especially large when only a small fraction of the workforce is directly affected by the minimum wage (as is true in the United States). The contribution of these omitted variables may be sizable compared with the relatively small expected effect of the minimum wage on aggregate employment. As a result, the bias arising from shocks to the upper tail can be particularly severe when we are interested in estimating the overall employment effect of the minimum wage.

There are numerous advantages of decomposing the aggregate employment changes by wage bins. First, such a decomposition allows us to focus on employment changes locally around the new minimum wage—the part of the wage distribution where we expect the policy to play a role. This variation is highly informative, yet rarely exploited. Second, and more importantly, our localized approach allows us to estimate the effects on overall employment as well as for subgroups where the standard approaches often fail to provide meaningful estimates on employment and wages. When only a small fraction of workers are directly affected by the minimum wage, the effect on the average wage of such subgroups will be very small. Without a clear wage effect, it is

5. This idea is similar to [Autor, Manning, and Smith \(2016\)](#) who use unrealistically large spillover effects to validate the empirical model in use.